Exotic Triangulations and the Octahedral Axiom

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(Co)chain complexes and (co)homology

Mathematicians are often interested in studying the cohomology of some cochain complexes over some abelian category $\mathscr{A}$ (e.g. of modules over some ring).

The category $\mathbf{C}(\mathscr{A})$ is also abelian, which is nice. In particular, we have the notion of short exact sequence, and a short exact sequence of complexes induces a long exact sequence in cohomology

$$\begin{tikzcd} \dots \arrow[r] & H^n(A) \arrow[r] & H^n(B) \arrow[r] & H^n(C) \arrow[dll, rounded corners] \\ & H^{n+1}(A) \arrow[r] & H^{n+1}(B) \arrow[r] & H^{n+1}(C) \arrow[r] & \dots \end{tikzcd}$$

(Co)chain complexes and (co)homology

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Homotopy category and derived category

Many constructions over morphisms of cochain complexes only depend on the homotopy class, so it makes sense to work in the (smaller) homotopy category

$$\mathbf{K}(\mathscr{A}) := \mathbf{C}(\mathscr{A})/\simeq .$$

Another standard construction is the derived category $\mathbf{D}(\mathscr{A})$, which inverts the homology isomorphisms.

However, neither of those categories is abelian. But they do have a triangulated structure!

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What is a triangulated category? I

Definition

A *pretriangulated category* is an additive category $\mathscr{T}$ together with an autoequivalence $\Sigma\colon \mathscr{T} \to \mathscr{T}$ and a class of distinguished triangles

$$X \to Y \to Z \to \Sigma X$$

satisfying the following axioms:

Definition

TR1 Every triangle which is isomorphic to a distinguished triangle is distinguished. Every triangle $0 \to X \xrightarrow{id} X \to 0$ is distinguished. Every morphism $\alpha : X \to Y$ can be completed to a distinguished triangle $X \to Y \to Z \to \Sigma X$.

TR2 A triangle $X \xrightarrow{\alpha} Y \xrightarrow{\beta} Z \xrightarrow{\gamma} \Sigma X$ is distinguished if and only if the triangle $Y \xrightarrow{\beta} Z \xrightarrow{\gamma} \Sigma X \xrightarrow{-\Sigma\alpha} \Sigma Y$ is distinguished.

TR3 Given a commutative diagram with distinguished rows, there exists a morphism $h \colon Z \to Z'$ making the whole diagram commute

$$\begin{tikzcd} X \arrow[r] \arrow[d, "f"] & Y \arrow[r] \arrow[d, "g"] & Z \arrow[r] \arrow[d, dotted, "h"] & \Sigma X \arrow[d, "\Sigma f"] \\ X' \arrow[r] & Y' \arrow[r] & Z' \arrow[r] & \Sigma X' \end{tikzcd}$$

The octahedral axiom

Definition

A *triangulated category* is a pretriangulated category satisfying the *octahedral axiom*:

TR4 Given distinguished triangles $(\alpha_1, \beta_1, \gamma_1)$, $(\alpha_2, \beta_2, \gamma_2)$ and $(\alpha_3, \beta_3, \gamma_3)$ such that $\alpha_3 = \alpha_2 \circ \alpha_1$, there exists a distinguished triangle $(\alpha_4, \beta_4, \gamma_4)$ such that the following diagram commutes

$$
\begin{tikzcd}
X \arrow[r, "\alpha_1"] \arrow[d, equal] & Y \arrow[r, "\beta_1"] \arrow[d, "\alpha_2"] & U \arrow[r, "\gamma_1"] \arrow[d, dotted, "\alpha_4"] & \Sigma X \arrow[d, equal] \\
X \arrow[r, "\alpha_3"] & Z \arrow[r, "\beta_3"] \arrow[d, "\beta_2"] & V \arrow[r, "\gamma_3"] \arrow[d, dotted, "\beta_4"] & \Sigma X \arrow[d, "\Sigma\alpha_1"] \\
 & W \arrow[r, equal] \arrow[d, "\gamma_3"] & W \arrow[r, "\gamma_3"] \arrow[d, dotted, "\gamma_4"] & \Sigma Y \\
 & \Sigma Y \arrow[r, "\Sigma\beta_1"'] & \Sigma U &
\end{tikzcd}
$$

Short exact sequences vs distinguished triangles

Distinguished triangles are the analogue of short exact sequences in the triangulated setting. For instance, a distinguished triangle in $\mathbf{K}(\mathscr{A})$ induces a long exact sequence in cohomology.

With this in mind, the octahedral axiom can be seen as an analogue of the second isomorphism theorem:

$$\begin{tikzcd}
A \arrow[r, hook] \arrow[d, equal] & B \arrow[r, two heads] \arrow[d, hook] & B/A \arrow[d, hook] \\
A \arrow[r, hook] & C \arrow[r, two heads] \arrow[d, two heads] & C/A \arrow[d, two heads] \\
& C/B \arrow[r, "\cong"] & \frac{C/A}{B/A}
\end{tikzcd}$$

Open problem: There is no known example of a pretriangulated category that does not satisfy the octahedral axiom, nor has it been proven to be a consequence of (TR1)-(TR3).

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Outline

Non-exotic = enhanceable

Derived categories are not the only widely used examples of triangulated categories. We could make the following division:

- **Algebraic:** those that are equivalent to the stable category of a Frobenius category $\underline{\mathscr{F}}$.
- **Topological:** those that are equivalent to the homotopy category of a stable model category $ho\mathscr{M}$.
- **Exotic:** those that are neither algebraic nor topological, i.e. those that admit no enhancement.

Exotic examples

There are only a few known examples of exotic triangulated categories:

- Muro, Schwede and Strickland [MSS07] gave an exotic structure to the category of finitely generated projective modules over $\mathbb{Z}/4\mathbb{Z}$.
- Rizzardo and Van den Bergh [RV20] gave a class of exotic examples which are linear over fields of characteristic zero.

We have found a third class of examples which are linear over algebraically closed fields of any characteristic.

Outline

A theorem by A. Heller

We first need a couple of previous results. The first one is a classical theorem on the subject.

Theorem (A. Heller, 1968, [Hel68])

Let $\mathscr{T}$ be an additive category with an autoequivalence Σ. Then, the set of pretriangulated structures on $(\mathscr{T}, \Sigma)$ is in bijection with the set of triangulated natural isomorphisms between Ω^{-3} and Σ, seen as triangulated functors $\underline{mod}(\mathscr{T}) \to \underline{mod}(\mathscr{T})$.

Previous work by Muro

The second one is in the more particular world of finite dimensional algebras.

Theorem (F. Muro, 2022, [Mur22])

Let Λ be a basic connected non-separable self-injective twisted-3-periodic f.d. algebra over a perfect field k. Then, there is an automorphism $\sigma \in Aut(\Lambda)$ such that twisting by σ is essentially the unique shift functor $\Sigma = (-)_\sigma \colon proj(\Lambda) \to proj(\Lambda)$ that makes $(proj(\Lambda), \Sigma)$ admit an algebraic triangulated structure.

Idea of the construction

The idea is to modify the automorphism σ into another automorphism $\tau = \sigma\kappa$ such that

$$(-)_\tau \ncong (-)_\sigma \colon \mathit{proj}(\Lambda) \to \mathit{proj}(\Lambda)$$

but $(-)_\tau \colon \underline{\mathit{mod}}(\Lambda) \to \underline{\mathit{mod}}(\Lambda)$ is still naturally isomorphic to Ω^{-3} as triangulated functors. In particular, this is the case if κ is a representative for a nontrivial element of the kernel of

$$\mathit{Pic}(\Lambda) \to \underline{\mathit{Pic}}(\Lambda),$$

where

$$\mathit{Pic}(\Lambda) = \mathit{Aut}(\Lambda)/\mathit{Inn}(\Lambda) \cong \{\text{ Self-equivalences of } \mathit{proj}(\Lambda) \ \} / \cong,$$

$$\underline{\mathit{Pic}}(\Lambda) := \{\text{ Triangulated self-equivalences of } \underline{\mathit{mod}}(\Lambda) \ \} / \cong_{tr} .$$

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Idea of the construction

The picture is as follows:

$$\begin{tikzcd}
\text{Alg. triang. str. on } (\mathit{proj}(\Lambda), (-)_\sigma) \arrow[r, "\text{Heller}"] & \delta \colon (-)_\sigma \cong \Omega^{-3} \arrow[d] \\
\text{Pretriang. str. on } (\mathit{proj}(\Lambda), (-)_\tau) & \varepsilon \colon (-)_\tau \cong \Omega^{-3} \arrow[l, "\text{Heller}"']
\end{tikzcd}$$

where ε is defined as a horizontal composition of natural isomorphisms

$$\begin{tikzcd}
\underline{\mathit{mod}}(\Lambda) \arrow[r, bend left, "(-)_\sigma"] \arrow[r, bend right, "\Omega^{-3}"'] \arrow[r, phantom, "\Downarrow \delta"] & \underline{\mathit{mod}}(\Lambda) \arrow[r, bend left, "(-)_\kappa"] \arrow[r, bend right, "\mathit{Id}"'] \arrow[r, phantom, "\Downarrow \eta"] & \underline{\mathit{mod}}(\Lambda)\,.
\end{tikzcd}$$

Our example

So the problem reduces to find an algebra Λ satisfying the conditions of the theorem in [Mur22] and such that the kernel of $Pic(\Lambda) \to \underline{Pic}(\Lambda)$ is nontrivial. We have found a family of examples among the deformed preprojective algebras of type L_2, which are given by the following quiver with relations:

$$P(L_2) \; = \; k\left(b \circlearrowright 1 \underset{a_2}{\overset{a_1}{\rightleftarrows}} 2 \right) \Big/ \left(b^2 + a_1 a_2, \; a_2 a_1\right).$$

The nontrivial element of the kernel is the class of the automorphism

$$\kappa \colon P(L_2) \to P(L_2), \quad a_1 \mapsto a_1, \; a_2 \mapsto a_2, \; b \mapsto b + b^3.$$

This example has the particularity that $(-)_\kappa$ is actually equal to the identity functor on $proj(P(L_2))$ as triangulated functors over $\underline{mod}(P(L_2))$, so we can take $\eta = id$.

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$$P(L_2) \;=\; k\left(\begin{tikzcd} b \circlearrowright 1 \arrow[r, shift left, "a_1"] & 2 \arrow[l, shift left, "a_2"] \end{tikzcd} \right) \Big/ \left(b^2 + a_1 a_2,\; a_2 a_1 \right).$$

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Our example

- It is now clear that the pretriangulated structure given by ε on $(proj(P(L_2)), (-)_\tau)$ is not algebraic, since $(-)_\tau \ncong (-)_\sigma$.
- Furthermore, it can also be shown to be non-topological.
- There is a last important thing to check: whether or not it satisfies the octahedral axiom. **Spoiler:** We don't know.

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- It is now clear that the pretriangulated structure given by ε on $(proj(P(L_2)), (-)_\tau)$ is not algebraic, since $(-)_\tau \not\cong (-)_\sigma$.
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Outline

Can't we disprove it?

If the octahedral axiom did not hold, it would be the first example of a pretriangulated category that is not triangulated, so we would really like that.

We have tried to find counterexamples by computer, but have not been able to find any "small" counterexample. Actually, every completion of the diagram that we try gives a distinguished cone.

Realistically, the most likely scenario is that it does hold.

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Realistically, the most likely scenario is that it does hold.

How can we prove it?

Strategies we have tried:

- Recover it from the octahedral axiom on the algebraic structure.
- Adapt the proof of the algebraic case.
- Use a (computable) sufficient condition given in [Mur20] in terms of the differential of a spectral sequence. It turns out that it is not satisfied. It is also the first example where this condition has been checked with a negative result.
- **Classifying initial data:** It turns out that the octahedral axiom does only depend on the induced stable morphism $ker(\alpha_1) \to \ker(\alpha_2)$, so we can try to classify all the possible initial data and check the axiom for each possible case. We don't know how to do this, but it may be a path to follow.

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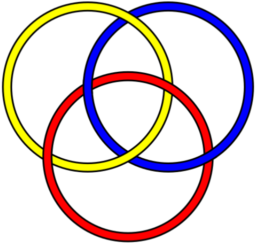
A new characterization

It is to be noticed that, until now, there was no known condition on the image of a pretriangulated structure under Heller's bijection that were equivalent to the octahedral axiom. Fernando had an intuition of how such a condition might look like and we have been able to prove his intuition was right.

The answer is given by a secondary cohomology operation. The first example of such operations is the Massey triple product.



Definition (Naturality defect product)

Let $F, G\colon \mathscr{A} \to \mathscr{B}$ be dg functors and $\tilde{\delta}$ a graded natural transformation from $H^*(F)$ to $H^*(G)$ of degree t. Moreover, let δ_X be a representative for $\tilde{\delta}_X$ for every X in $\mathscr{A}$. Suppose $a = [\alpha] \in H^p(\mathscr{A})(A, B)$ and $b = [\beta] \in H^q(\mathscr{A})(B, C)$ have vanishing composition. More precisely, let

$$\beta\alpha = d(s), \qquad D_\delta(\alpha) = d(x), \qquad D_\delta(\beta) = d(y).$$

Then, we define the *naturality defect product* on a and b with respect to $\tilde{\delta}$ by

$$\langle b, a\rangle_{\tilde{\delta}} := \big[\,[\,D_\delta(s) - (-1)^t\big(yF(\alpha) + (-1)^{q+tq}G(\beta)x\big)\,]\,\big],$$

which is seen as an element of the quotient

$$\frac{H^{p+q+t-1}(\mathscr{B})(F(A), G(C))}{G(b)\circ H^{p+t-1}(\mathscr{B})(F(A), G(B)) + H^{q+t-1}(\mathscr{B})(F(B), G(C))\circ F(a)}.$$

Naturality defect product

The picture is as follows:

$$
\begin{tikzcd}
F(A) \arrow[r, "F(\alpha)"] \arrow[d, "\delta_A"'] \arrow[dr, Rightarrow, "x"] \arrow[rr, bend left=40, "0"] \arrow[rr, bend left=20, phantom, "\Uparrow F(s)"] & F(B) \arrow[r, "F(\beta)"] \arrow[d, "\delta_B"] \arrow[dr, Rightarrow, "y"] & F(C) \arrow[d, "\delta_C"] \\
G(A) \arrow[r, "G(\alpha)"'] \arrow[rr, bend right=40, "0"'] \arrow[rr, bend right=20, phantom, "\Downarrow G(s)"] & G(B) \arrow[r, "G(\beta)"'] & G(C)
\end{tikzcd}
$$

The characterization

In the triangulated setting, we take all dg categories to be the dg category $\mathscr{A}$ of acyclic complexes over $\mathscr{T}$, so that $H^0(\mathscr{A}) = \underline{Ac}(\mathscr{T}) \simeq \underline{mod}(\mathscr{T})$.

Theorem (DC.-Muro, 2026)

Let $\mathscr{T}$ be a pretriangulated category with associated natural transformation $\tilde{\delta}$. Then, the following are equivalent:

1. *$\mathscr{T}$ is triangulated.*
2. *The naturality defect product $\langle g, f\rangle_{\tilde{\delta}}$ vanishes for every morphism f between exact triangles and every morphism g such that $gf = 0$.*

Does this perspective help for our example? It is also unclear.

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Thank you for your attention!